FROM TRANSPARENT TO OPAQUE: LEVERAGING GATING MECHANISM IN LATENT REASONING CHAIN

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ABSTRACT

We proposed a novel paradigm for implementing latent reasoning by combining DAPO-like reinforcement learning and special gating units. In this way, we transformed text-based explicit reasoning models to latent reasoning models, progressively and practically. Initial experiments demonstrate its feasibility, with significant stability and the least extra training overhead by parallelism and minimal changes, while holding full potential to scale up.

1 INTRODUCTION

The Chain of Continuous Thought (COCONUT) (Hao et al., 2024) explores the potential of the latent layer of reasoning as an alternative to the traditional explicit CoT (Wei et al., 2022) in both the reasoning procedure and the response procedure. Its previous work has shown that the latent layer of continuous reasoning can reduce unnecessary statements and focus on the core of the problem.

These models are issued from general-purpose language, trained in supervised steps of a reasoning procedure, and progressively increase their layers of latent reasoning. It is impractical due to the lack of customized reasoning data, and progressive training is significantly less convenient than explicit CoT. Hence, we provide a new approach to implementing this latent reasoning layer, fine-tuning from explicit reasoning models.

In 2014 (Chung et al., 2014), Chung et al. proposed the Gated Recurrent Unit. It is first leveraged to improve the RNNs, but we found that using it in the latent reasoning chain is also feasible to maintain its long-term contextual awareness.

In January, the DeepSeek team proposed a distillation-based method to implement a reasoning model through reinforcement learning (DeepSeek-AI, 2025), and applied the approach to Qwen-2.5B (Yang et al., 2024) to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement COCONUT on the small model and find its potential in reasoning compared to the original explicit reasoning model.

2 RELATED WORK

3 METHODOLOGY

3.1 ARCHITECTURE

Training a language model from direct answering to latent reasoning is impractical without specialized guidance. Instead, we propose a direct method, alternating the model architecture from the pure connection of latent reasoning layers to a GRU-based latent reasoning chain.

We deploy a reset gate and an update gate, not during the model architecture but in the reasoning process.

3.1.1 LANGUAGE MODEL

We use the architecture proposed in COCONUT (Hao et al., 2024). It enables two modes for LLMs: latent mode and transparent mode.

Latent Mode The latent mode reduces the extra computation of decoding reasoning text. Hao et al. demonstrate that human language is designed to communicate instead of relying heavily on reasoning, so we believe it's unnecessary for users always to decipher what AI reasons. This mode also exclusively owns the gated unit, which will be explained later.

Transparent Mode This is the normal mode of LLMs. They speak the human language and eventually provide answers during our research progress.

3.1.2 GATED UNIT

The gated unit is the innovation point of our paper.

Unlike standard causal models that treat each decoding step as conditionally independent beyond the token history, we introduce a GRU-style gating mechanism in the latent reasoning stage. This gate learns to control information flow and retention, enabling the model to persist and refine internal reasoning states across steps. This effectively transforms reasoning from a fully recurrent process into a gated latent update, yielding better consistency and longer-term dependency tracking.

3.2 REINFORCEMENT LEARNING

Inspired by DAPO Yu et al. (2025), we deployed a mechanism that applies standardization considered beyond accuracy.

3.2.1 SAMPLING

The key to letting the model learn is to sample and filter satisfying results.

The whole sampling procedure is shown in figure 2. We will provide our hyperparameters in the Appendix.

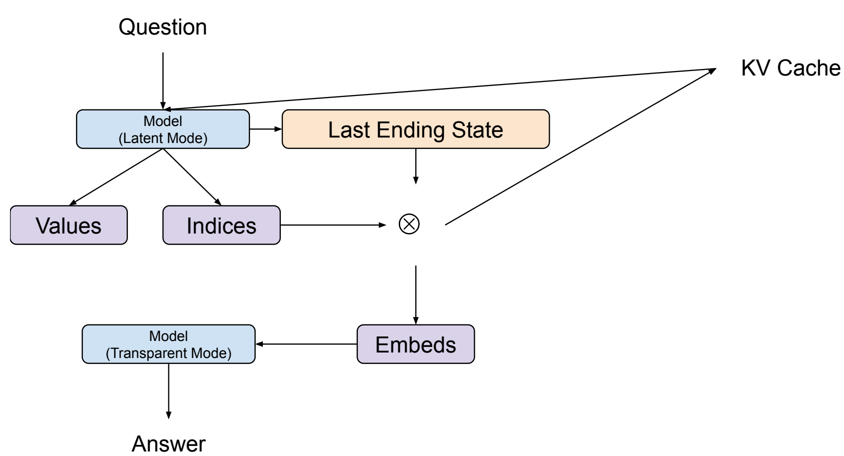

Figure 1: The sampling procedure. It happens with parallelized computation, but for the sake of simplicity, we use one as an example.

3.2.2 AWARD MECHANISM

DeepSeek's method unintentionally increased the reasoning procedure to an abnormal length.

Since it is batch, the final formula is given in 1, where r_i, c_i, and l_i denote the award, correctness award, and length award, respectively.

$$r_i = \frac{c_i - \bar{c}}{\sqrt{\sum (c_i - \bar{c})^2} + 10^{-6}} + \frac{l_i - \bar{l}}{\left|\max\left\{l_i - \bar{l}\right\}\right| + 10^{-6}} \tag{1}$$

By adding 10^{-6}, we can avoid producing `NaN` if dividing zero. Then, we apply these scores to the loss, given by 2. We are also leveraging gradient accumulation during the training procedure.

$$\text{loss} = \sum \text{model loss} + \frac{\overline{r_i}}{\sum \text{text end indices} + 1} \tag{2}$$

And to promote its diversity, we are also applying punishment for being identical.

A Experiments

B Discussion

C Conclusion

References

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D Appendix